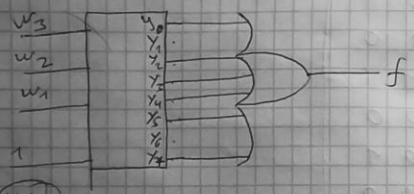
4.1

$f(w_1, w_2, w_3) = \sum m(0, 2, 3, 4, 5, 7)$

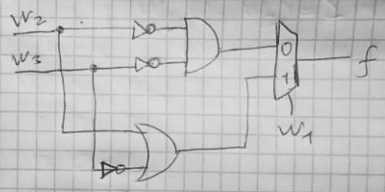
$f = \bar{w}_1 \bar{w}_2 \bar{w}_3 + \bar{w}_1 w_2 \bar{w}_3 + \bar{w}_1 w_2 w_3 + w_1 \bar{w}_2 \bar{w}_3 + w_1 \bar{w}_2 w_3 + w_1 w_2 \bar{w}_3$



4.4

$f = \bar{w}_2 \bar{w}_3 + w_1 w_2$

w_1	w_2	w_3	f	
0	0	0	1	
0	0	1	0	$\bar{w}_2 \bar{w}_3$
0	1	0	0	
0	1	1	0	
1	0	0	1	
1	0	1	0	$w_2 + \bar{w}_3$
1	1	0	1	
1	1	1	1	



4.7

$f = \bar{w}_2 + \bar{w}_1 \bar{w}_3 + w_1 w_3 = \bar{w}_1(\bar{w}_2 + 0 w_3 + 0 w_3) +$
$+ w_1(\bar{w}_2 + 1 \bar{w}_3 + 1 w_3) = \bar{w}_1(\bar{w}_2 + \bar{w}_3) + w_1(\bar{w}_2 + w_3) =$
$= \bar{w}_1 \bar{w}_2 + \bar{w}_1 \bar{w}_3 + w_1 \bar{w}_2 + w_1 w_3 =$
$= \bar{w}_2(\bar{w}_1 0 + \bar{w}_1 \bar{w}_3 + w_1 0 + w_1 w_3) + w_2(\bar{w}_1 1 + \bar{w}_1 \bar{w}_3 +$
$+ w_1 1 + w_1 w_3) = \bar{w}_2(\bar{w}_1 + \bar{w}_1 \bar{w}_3 + w_1 + w_1 w_3) +$
$+ w_2(\bar{w}_1 \bar{w}_3 + w_1 w_3) = \bar{w}_1 \bar{w}_2 + \bar{w}_1 \bar{w}_2 \bar{w}_3 + w_1 \bar{w}_2 +$
$+ w_1 \bar{w}_2 w_3 + \bar{w}_1 w_2 \bar{w}_3 + w_1 w_2 w_3 = \bar{w}_3(\bar{w}_1 \bar{w}_2 +$
$+ \bar{w}_1 \bar{w}_2 0 + w_1 \bar{w}_2 + w_1 \bar{w}_2 0 + \bar{w}_1 w_2 0 + w_1 w_2 0) +$
$+ w_3(\bar{w}_1 \bar{w}_2 + \bar{w}_1 \bar{w}_2 1 + w_1 \bar{w}_2 + w_1 \bar{w}_2 \cdot 1 + \bar{w}_1 w_2 1 +$
$+ w_1 w_2 \cdot 1) = \bar{w}_1 \bar{w}_2 \bar{w}_3 + \bar{w}_1 \bar{w}_2 \bar{w}_3 + w_1 \bar{w}_2 \bar{w}_3 +$
$+ \bar{w}_1 w_2 \bar{w}_3 + \bar{w}_1 \bar{w}_2 w_3 + w_1 \bar{w}_2 w_3 + w_1 w_2 w_3 +$
$+ w_1 w_2 w_3 = \bar{w}_1 \bar{w}_2 \bar{w}_3 + w_1 \bar{w}_2 \bar{w}_3 + \bar{w}_1 w_2 \bar{w}_3 +$
$+ \bar{w}_1 \bar{w}_2 w_3 + w_1 \bar{w}_2 w_3 + w_1 w_2 w_3$

$f(w_1, w_2, w_3) = \sum m(0, 1, 2, 4, 5, 7)$